

Validity of the Kinarm Approach-Avoidance Task for Assessing Automatic Tendencies Towards Physical Activity and Sedentary Behaviours: A Registered Report

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ABSTRACT

Background. Approach-avoidance tendencies have traditionally been assessed using unidimensional reaction times, measured by moving an on-screen manikin towards or away from stimuli using a keyboard, joystick, touchscreen, or mouse. Recently, a bimanual robotic task, referred to as the Kinarm Approach-Avoidance Task (KAAT), has been proposed to provide a continuous, high-resolution, multidirectional, and ecologically relevant assessment of the spatiotemporal dynamics that characterize these tendencies.

Objective. To assess the construct, concurrent, convergent, and predictive validity of the KAAT with stimuli depicting physical activity and sedentary behaviours.

Methods. Healthy participants performed the KAAT and the keyboard-based manikin task. Approach-avoidance tendencies were quantified using reaction time, time to peak velocity, x-flips, and opposite displacement. Physical activity was assessed with accelerometers. Additional questionnaires measured intentions, explicit attitudes, automaticity, and the valuation of physical effort. Trial-level approach-avoidance tendencies were analysed using linear mixed-effects models. Concurrent validity between the KAAT and the manikin task and predictive associations with physical activity and related psychological constructs were evaluated using correlations and hierarchical regressions, controlling for age, sex, height, weight, handedness, error rate, and elapsed time.

Keywords. Psychology, Motivation, Robotics, Validation Study, Exercise, Sedentary Behaviour

INTRODUCTION

In recent years, dual-process models have been proposed to explain physical activity behaviour [1–4]. These models suggest that physical activity is shaped by both controlled (also called reflective) processes (e.g., explicit attitudes, intentions, valuation of physical effort) and automatic processes (e.g., automatic affective reactions, approach-avoidance tendencies). Controlled processes are deliberative, resource-intensive, and conscious, whereas automatic processes are fast, effortless, and operate with limited conscious awareness [5]. According to these models, repeated experiences of pleasure or displeasure during physical activity create learned associations that generate automatic affective evaluations when physical activity-related cues are encountered. These evaluations can influence controlled processes and elicit approach or avoidance impulses that affect the likelihood of being physically active. Among automatic processes, approach-avoidance tendencies are thought to be particularly proximal to behaviour, as they reflect automatic action orientations towards or away from environmental stimuli [6–9].

In health psychology, approach-avoidance tendencies have traditionally been assessed using unidimensional reaction times to task-relevant stimuli, measured via interfaces such as a keyboard, joystick, touchscreen, or computer mouse [10–14]. Consistent with this broader literature, physical activity research has primarily assessed approach-avoidance tendencies using the manikin task in which reaction times are measured by moving an on-screen manikin towards (i.e., approach) or away (i.e., avoid) from stimuli depicting physical activity and sedentary behaviours [9,15–32]. By measuring the time required to initiate conceptual (i.e., via keyboard) or actual (e.g., via joystick) approach or avoidance movements in response to stimuli related to physical activity, reaction times provide an indirect measure of the strength and direction of automatically activated tendencies.

Despite their theoretical grounding and widespread use, unidimensional reaction-time measures have limitations. Because reaction times reduce the motor response to a single timepoint, they conflate multiple perceptual, cognitive, and motor processes while discarding kinematic information about movement execution. As a result, they fail to capture the temporal dynamics and spatial characteristics of behaviour, such as trajectory, amplitude, and changes in direction. In addition, constraining responses to a single spatial dimension (e.g., anteroposterior joystick movements) may oversimplify the inherently multidirectional nature of approach-avoidance tendencies and limit sensitivity to how these tendencies are expressed across space.

Other limitations relate to the task implementation. In joystick-based tasks, approach (i.e., pulling) and avoidance (i.e., pushing) are often simulated via stimulus zooming in and out, respectively, which introduces ambiguity because the same response can be interpreted differently. For example, pulling a joystick brings the arm away from the screen and stimulus (i.e., withdrawal), yet it is typically paired with visual stimulus enlargement, which conveys approach. This mismatch may confound the interpretation of reaction-time differences [23]. By relying on symbolic and discretized responses, keyboard-based tasks such as the manikin task avoid this reference-frame ambiguity. However, because these key presses only indirectly reflect bodily action, these tasks lack ecological validity and sensitivity to fine-grained temporal and spatial aspects of movement execution.

Taken together, these limitations highlight the need for paradigms that capture approach-avoidance tendencies as a dynamic and embodied process rather than as a single latency measure constrained to one spatial dimension. To address this need, a bimanual robotic task implemented on the Kinarm platform has been proposed that provides a continuous, high-resolution, multidirectional, and ecologically relevant assessment of the spatiotemporal dynamics that

characterize approach-avoidance tendencies [33]. This so-called Kinarm Approach-Avoidance Task (KAAT) records kinematic data from two handles in a two-dimensional virtual environment, enabling the quantification of movement patterns that may reflect approach-avoidance conflicts, movement-related uncertainty, and hesitation beyond what can be assessed with classical reaction-time-based tasks. However, this task has not yet undergone formal psychometric validation. Accordingly, the objective of this study is to evaluate the construct, convergent, concurrent, and predictive validity of the KAAT.

Hypotheses

We formulated hypotheses (H) corresponding to each of the four types of validity addressed by our objective (Table 1). Raw bias refers to the absolute tendency to approach or avoid a stimulus, whereas relative bias refers to the relative tendency, which accounts for interindividual differences in generic approach-avoidance tendencies by controlling for responses to neutral stimuli [9]. Approach-avoidance tendencies were derived from these two types of bias, each calculated from the four dependent variables: reaction time, time to peak velocity, x-flips, and opposite displacement.

Construct Validity

To assess the structural aspect of construct validity, which examines whether a measure follows expected internal patterns, we compared raw and relative bias scores separately against zero to evaluate whether the measure could detect systematic bias. We hypothesised that, for both physical activity and sedentary stimuli, KAAT-derived raw bias (H1a–h) and relative bias (H1i–p) would significantly differ from zero across all outcome variables (i.e., reaction time, time to peak velocity, x-flips, and opposite displacement).

As an additional examination of construct validity, we compared the magnitude of reaction time-based approach-avoidance effects between the KAAT and the manikin task to examine whether KAAT-derived bias behaved as theoretically expected relative to an established task. We hypothesised that raw bias would be larger when assessed with the KAAT than with the manikin task for both physical activity (H1q) and sedentary stimuli (H1r). We formulated analogous hypotheses for relative bias (H1s, H1t).

Concurrent Validity

To assess concurrent validity, which examines the extent to which a new measure is associated with an established criterion assessed at the same time [34], we examined associations between raw and relative bias score derived from both tasks. We hypothesised that raw bias would show small positive correlations between tasks for both physical activity (H2a) and sedentary stimuli (H2b), consistent with previous studies showing limited convergence between approach-avoidance tasks [11,35]. We formulated analogous hypotheses for relative bias (H2c, H2d).

Convergent Validity

To assess convergent validity, which examines the extent to which a measure is associated with other measures of theoretically related constructs [34], we examined the association between KAAT-derived relative bias and six outcomes and compared them with the corresponding associations obtained using the manikin task. We hypothesised that the relative bias towards physical activity derived from the KAAT would be associated with (H3a) device-based total physical activity, (H3b) device-based moderate-to-vigorous physical activity, (H3c) explicit

affective attitudes towards physical activity, (H3d) intentions to be physically active, (H3e) valuation of physical effort, and (H3f) automaticity towards physical activity. We further hypothesised that these associations would be stronger for the KAAT than for the manikin task (H3g–l).

Predictive Validity

To assess predictive validity, which refers to the extent to which a measure predicts relevant behavioural outcomes [34], we computed the variance in accelerometer-based total physical activity explained by KAAT-derived and manikin-derived relative bias scores. We hypothesised that reaction-time-based relative bias towards physical activity stimuli measured with the KAAT would explain variance in device-based total physical activity (H4a), and that adding kinematic-based relative biases (i.e., time to peak velocity, x-flips, and opposite displacement) would explain additional variance (H4b). We further hypothesised that the reaction-time-based relative bias towards physical activity stimuli measured with the KAAT would explain more variance than when assessed with the manikin task (H4c).

Table 1. A priori hypotheses by validity type

Construct Validity (H1)		
<i>Internal Structure</i>		
H1a: KAAT raw bias in RT _{PA} ≠ 0	H1i: KAAT relative bias in RT _{PA} ≠ 0	
H1b: KAAT raw bias in RT _{SED} ≠ 0	H1j: KAAT relative bias in RT _{SED} ≠ 0	
H1c: KAAT raw bias in time to peak velocity _{PA} ≠ 0	H1k: KAAT relative bias in time to peak velocity _{PA} ≠ 0	
H1d: KAAT raw bias in time to peak velocity _{SED} ≠ 0	H1l: KAAT relative bias in time to peak velocity _{SED} ≠ 0	
H1e: KAAT raw bias in x-flip _{PA} ≠ 0	H1m: KAAT relative bias in x-flip _{PA} ≠ 0	
H1f: KAAT raw bias in x-flip _{SED} ≠ 0	H1n: KAAT relative bias in x-flip _{SED} ≠ 0	
H1g: KAAT raw bias in opposite displacement _{PA} ≠ 0	H1o: KAAT relative bias in opposite displacement _{PA} ≠ 0	
H1h: KAAT raw bias in opposite displacement _{SED} ≠ 0	H1p: KAAT relative bias in opposite displacement _{SED} ≠ 0	
<i>Task-level differences in effect magnitude</i>		
H1q: KAAT raw bias in RT _{PA} > Manikin raw bias in RT _{PA}	H1s: KAAT relative bias in RT _{PA} > Manikin relative bias in RT _{PA}	
H1r: KAAT raw bias in RT _{SED} > Manikin raw bias in RT _{SED}	H1t: KAAT relative bias in RT _{SED} > Manikin relative bias in RT _{SED}	
Concurrent Validity (H2)		
H2a: KAAT & Manikin raw bias in RT _{PA} are associated	H2c: KAAT & Manikin relative bias in RT _{PA} are associated	
H2b: KAAT & Manikin raw bias in RT _{SED} are associated	H2d: KAAT & Manikin relative bias in RT _{SED} are associated	
Convergent Validity (H3)		
KAAT relative bias in RT _{PA} is associated with:		
H3a: Device-based total PA	H3c: Explicit attitudes towards PA	H3e: Valuation of physical effort
H3b: Device-based moderate-to-vigorous PA	H3d: Intentions to be PA	H3f: Automaticity towards PA
KAAT relative bias in RT _{PA} is more strongly associated than Manikin relative bias in RT _{PA} with:		
H3g: Device-based total PA	H3i: Explicit attitudes towards PA	H3k: Valuation of physical effort
H3h: Device-based moderate-to-vigorous PA	H3j: Intentions to be PA	H3l: Automaticity towards PA
Predictive Validity (H4)		
H4a: KAAT relative bias in RT _{PA} explains variance in device-based total PA		
H4b: Kinematic-based relative biases _{PA} explain additional variance beyond that accounted for by KAAT relative bias in RT _{PA}		
H4c: Relative bias in RT _{PA} better explains total PA when assessed with the KAAT than with the manikin task		

Notes. KAAT = Kinarm approach avoidance task, RT = reaction time, H = hypothesis, PA = physical activity, SED = sedentary behaviour

METHODS

Participants

Participants were recruited via social media, emails, posters, and word of mouth. Inclusion criteria were age between 18 and 60 years and the ability to communicate fluently in French or English. Exclusion criteria included any conditions that could influence our results, such as self-reported depression, upper or lower limb injury, a history of psychiatric or neurological disorders,

alcohol or drug abuse, or the use of psychotropic medications or illicit substances that may affect attention or neurological functioning. The study was approved by the University of Ottawa Research Ethics Board (H-01-25-11075). All participants provided informed consent and were not compensated for their participation. Data were collected between April 2025 and February 2026. Although 100% of the data have been collected, no observations, preprocessing, or analyses have been or will be conducted prior to stage 1 in-principle acceptance, which corresponds to Level 3 of bias control according to the Peer Community In Registered Report taxonomy.

A Priori Power Analysis

Simulation-Based Power Analysis for Linear Mixed-Effects Models

Our study was powered to detect a two-way interaction effect using Monte-Carlo simulations with the *simr* package in R [36], which is specifically designed for power estimation in mixed-effects models. Because no directly comparable effect size estimates were available in the literature, we derived the model parameters from an open dataset [37] obtained from a prior study [16] that used the same manikin task structure and stimuli as in our study. The data were processed and analysed using the same procedures planned for the present study. The only difference was that the following linear mixed-effects model was fitted separately for physical activity and sedentary stimuli to estimate the sample size required for each stimulus category: Reaction Time ~ Movement × Stimulus Type + (1 + Movement | Subject) + (1 | Pictogram). Movement reflected the response direction (approach vs. avoidance), whereas Stimulus Type contrasted physical activity with neutral pictograms in one model and sedentary with neutral pictograms in the other. Fixed-effect estimates, random-effect variance components (VarCorr), and residual variance (σ^2) from these pilot models were then used to generate simulated datasets reproducing the planned balanced experimental design for each experimental stimulus category. Statistical power for the Movement × Stimulus Type interaction was estimated using 1,000 Monte-Carlo simulations ($\alpha = .05$; t-test with Satterthwaite-approximated degrees of freedom on the fixed effect), yielding a Monte-Carlo standard error below ~1% for the power values around .90 [38]. We computed simulation-based power curves using the *powerCurve* function. These curves indicated that achieving 90% power required 32 participants for physical activity stimuli and 45 participants for sedentary stimuli (Figure 1). We therefore set 45 participants as the minimum required sample size. However, because the simulation-based power analysis relies on variance components and fixed-effect estimates derived from a single prior dataset and thus carries parameter uncertainty [39], recruitment continued until the pre-specified end of data collection in February 2026.

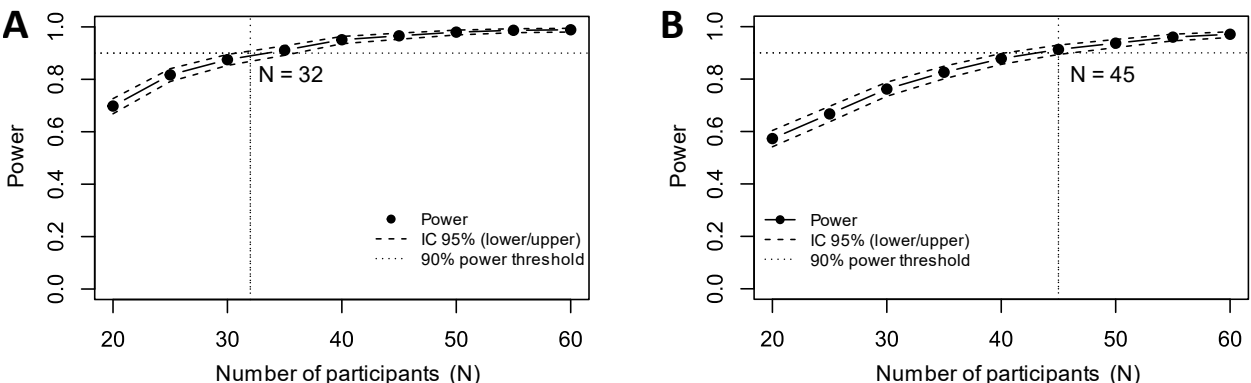


Figure 1. Simulation-based power curves for the interaction between Movement (approach vs. avoidance) and Stimulus Type (experimental vs. neutral) for physical activity (**A**) and sedentary stimuli (**B**). Note: IC 95% = 95% confidence interval.

Sensitivity Power Analyses

Sensitivity power analyses were conducted using the final sample size ($N = 96$) to estimate the minimum detectable effect size for each hypothesis [39], assuming a two-sided alpha level of .05 and a conventional target power of 80%. By identifying the range of effect sizes that the final sample is adequately powered to detect, these analyses provide context for interpreting both statistically significant and non-significant findings.

For H2a–d, the sensitivity power analysis was conducted using G*Power for a bivariate correlation exact test, and indicated a minimum detectable correlation of $|r| = .28$.

For H3, we estimated separate linear regression models for each theoretically related outcome variable. We conducted sensitivity power analyses in G*Power (F tests; linear multiple regression: fixed model, R^2 increase) to approximate the minimum detectable regression coefficient (for H3a–f). We did not estimate a minimum detectable coefficient difference (H3g–l), because G*Power cannot provide a reliable effect size for this contrast. As the G*Power framework yield a Cohen's f^2 estimate, we computed the corresponding additional explained variance (ΔR^2) and used it to derive the semi-partial correlation, which is of similar magnitude to the standardised regression coefficient (β) when multicollinearity is limited.

For H3a–f, the regression models included eight predictors: age, sex, height, weight, handedness, total number of errors, time on the task, and the KAAT relative bias term, with the latter being the sole predictor of interest. The smallest detectable incremental effect was $f^2 = 0.084$ for one tested predictor. This corresponds to approximately $\Delta R^2 = 0.05$ – 0.08 , when assuming plausible full-model R^2 values between 0.10 and 0.40, yielding a detectable semi-partial correlation of approximately $r = 0.23$ – 0.28 . Accordingly, without significant multicollinearity, $|\beta_{KAAT}|$ regression coefficients of this magnitude would be expected to be detectable with adequate statistical power.

For H4a–b, similar G*Power sensitivity power analysis parameters used for H3a–f were applied to determine the smallest detectable increments in explained variance (ΔR^2) for the sequential linear regression models. For H4a, both the predictor of interest and the total number of predictors were identical to those in H3a–f, yielding a minimal detectable incremental effect size of $f^2 = 0.084$. For H4b, three predictors (x-flips, opposite displacement, and time to peak velocity) were entered at the final step of the model, resulting in 11 total predictors. The minimum detectable incremental effect size was $f^2 = 0.119$. Expressed in terms of explained variance, these values correspond to detectable increases in ΔR^2 of approximately 5–8% for H4a and 7–11% for H4b.

Because testing H4c requires comparing non-nested models using information criteria, standard a priori or sensitivity power analyses based on null-hypothesis significance testing are not applicable.

Study Procedure

Participants performed the KAAT on the Kinarm Endpoint robotic platform and the manikin task on a laptop using Inquisit 6 software (Millisecond Software, 2015). Task order was

randomised, with a five-minute rest period between tasks. Questionnaires were administered after both tasks were completed, and collected information on age, sex, gender, weight, height, explicit attitudes towards physical activity, intention to be physically active, physical effort valuation, usual level of physical activity, and automaticity towards physical activity (Figure 2). One attention-check question was included in the questionnaires (“Please answer “2” to this question that allows us to verify that you actually read the questions”). At the end of the session, participants were provided with an ActiGraph wGT3X-BT accelerometer to record their physical activity over the subsequent 7 days.

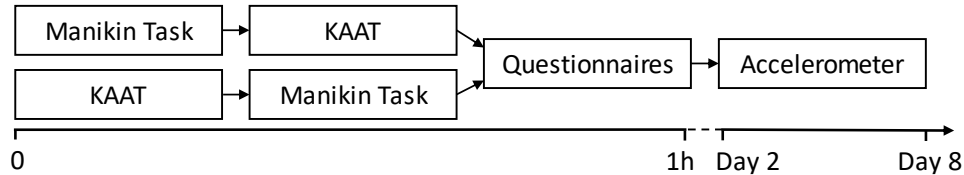


Figure 2. Study timeline. Note: KAAT = Kinarm approach-avoidance task

Approach-Avoidance Tasks

Kinarm Approach-Avoidance Task (KAAT)

The Kinarm Endpoint robotic platform is equipped with a 2-dimensional augmented reality display and provides high-resolution kinematic data. In the KAAT, participants stood while holding the robot’s handles, with each hand represented on the display by a white cursor (Figure 3A).

The KAAT began with on-screen instructions illustrating the upcoming condition. To initiate each block of trials, participants used either hand to place the white cursor into a green circle located at the centre of the workspace. For each trial, a blue circle then appeared either to the left or the right side of the display with equal probability. Once the participant placed the white cursor of the appropriate hand (i.e., blue circle in the left side required the left-hand cursor) into the blue circle and held it there for 500 to 750 ms, a stimulus target appeared at one of twelve possible locations positioned 10 cm from the start target and separated by 30-degree increments (Figure 3B). Simultaneously, a second target consisting of an open white circle appeared at the opposite location.

Participants were instructed that bringing the white cursor to the stimulus target corresponded to an “approach” response, whereas bringing the cursor to the opposite target corresponded to an “avoid” response. If neither target was reached within 3,000 ms, a “Too Slow!” message was displayed for 750 ms and the trial ended. If the participant incorrectly approached or avoided a target, an “Incorrect!” text was displayed for 750 ms before the trial ended (Figure 3C) [33]. A demonstration video of the KAAT is available on GitHub [40].

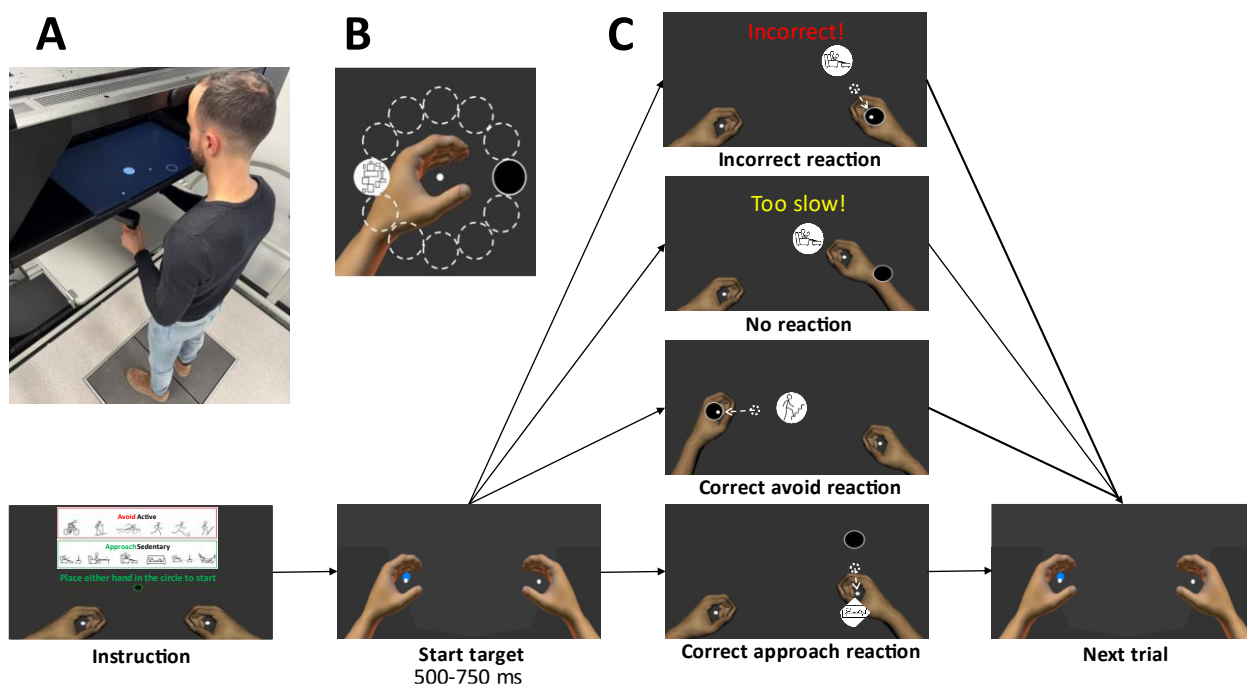


Figure 3. Kinarm Approach-Avoidance Task (KAAT). (A) Experimental setup. (B) Example trial in which a neutral stimulus appears in the west position and the open circle appears at the opposite location. Dashed circles illustrate the possible stimulus locations for visualisation purposes only; they are not visible to participants during the task. (C) Example trial in which the participant is instructed to approach sedentary stimuli and avoid physical activity stimuli. Hands are shown for illustrative purposes only and are not visible to participants during the actual task.

Manikin Task

The manikin task was implemented in Inquisit 6 software [9,18,28]. Participants performed the task in a standing position with the laptop in front of them, with one index finger on the “U” key and the other on the “N” key (Figure 4A). Pressing the “U” key moved the avatar upward, whereas pressing the “N” key moved the avatar downward. When the avatar appeared below the stimulus, the “U” key was associated with an approach movement, and the “N” key was associated with an avoidance movement. The opposite mapping applied when the avatar appeared above the stimulus.

The manikin task began with on-screen textual instructions indicating the upcoming condition. Each trial then started with a fixation cross presented at the centre of the screen for a random time ranging from 500 to 750 ms (Figure 4B). An avatar subsequently appeared at either the top or the bottom of the screen for 1000 ms, after which a stimulus was presented at the centre of the screen. Pressing the correct key triggered the onset of the next trial. If a participant pressed the incorrect key (e.g., pressing “U” when “N” was required), an “ERROR” message was displayed for 800 ms and the trial ended. If no key was pressed within seven seconds, a “TOO SLOW” message was displayed for 800 ms before the trial ended. A demonstration video of the manikin task is available on GitHub [40].

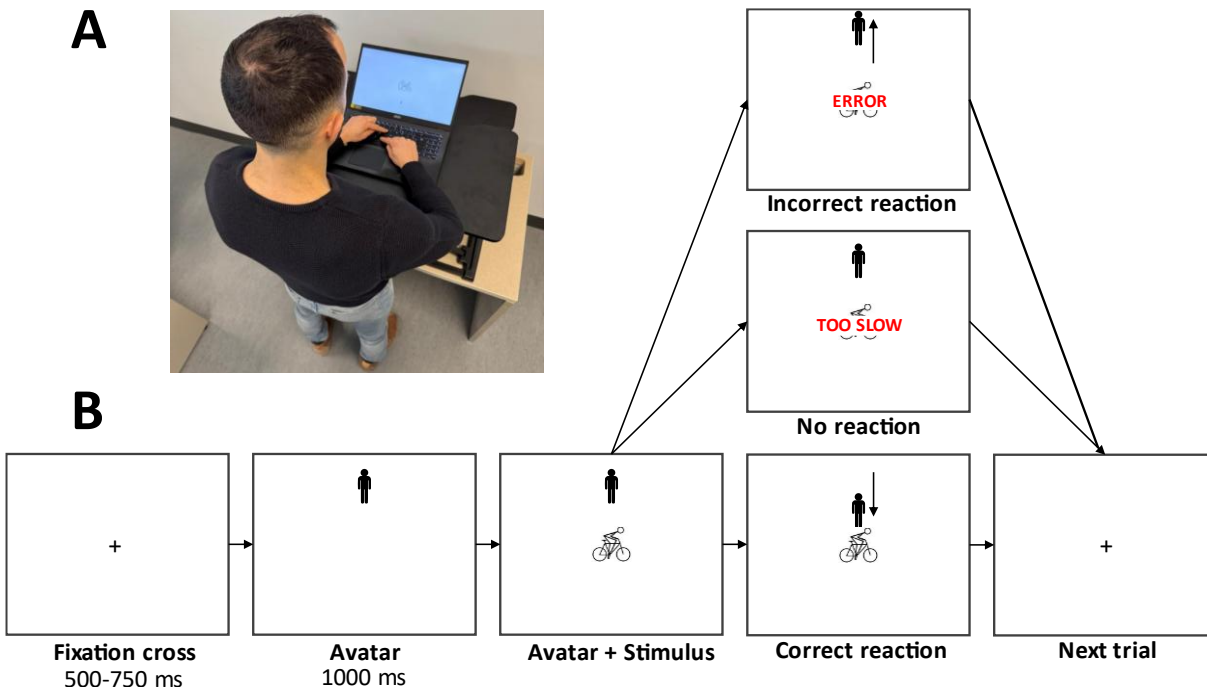


Figure 4. Manikin Task. (A) Experimental setup for the manikin task. (B) Example trial of the manikin task in which the participant is instructed to approach physical activity stimuli.

Experimental Conditions

In both the KAAT and the manikin task, task performance was recorded in two experimental conditions and two neutral conditions (Figure 5). Neutral trials were used to compute relative bias (see *Raw and Relative Automatic Approach-Avoidance Bias* section) [9].

In one experimental condition, participants were instructed to approach pictograms depicting physical activity and to avoid pictograms depicting sedentary behaviour. In the other experimental condition, the mapping was reversed: participants avoided physical activity stimuli and approached sedentary stimuli.

In the neutral conditions, experimental pictograms were replaced by stimuli with geometric shapes (circles or squares) matched in number and size of information to three physical activity pictograms (swimming, hiking, cycling) and three sedentary pictograms (couch, hammock, reading) [9,18,28,33]. In one neutral condition, participants were instructed to approach circles and avoid squares. In the other condition, the mapping was reversed.

One neutral block was administered at the beginning of the task and the other at the end. The two experimental blocks were completed between the neutral blocks. Each block included 48 trials, resulting in a total of 192 trials. Approach-avoidance response mapping was counterbalanced across participants separately for neutral (circles vs. squares) and experimental blocks (physical activity vs. sedentary behaviour). Within each block, pictogram presentation order was randomised across both types of stimuli.

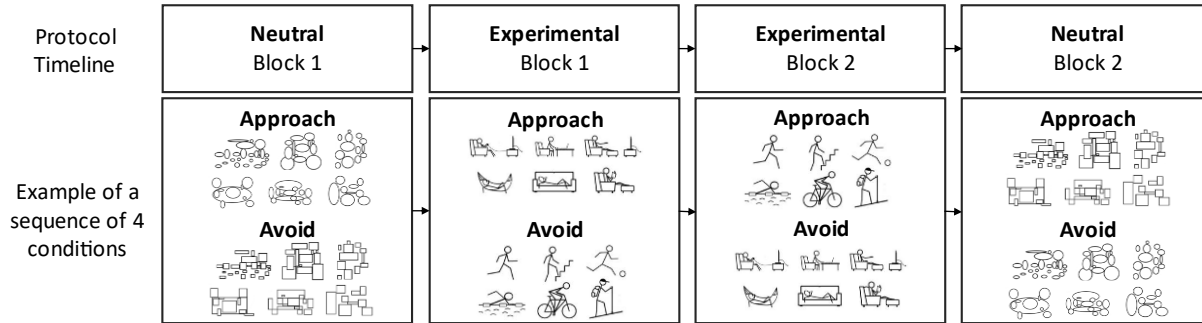


Figure 5. Timeline and stimuli used in the manikin task and the Kinarm Approach-Avoidance Task (KAAT). Approach-avoidance mapping was counterbalanced for participants separately for neutral (circles vs. squares) and experimental blocks (physical activity vs. sedentary behaviours). Within each block, pictogram presentation order was randomised across both types of stimuli.

Reaction Times

Incorrect responses, responses faster than 150 ms, and responses slower than 1,500 ms will be excluded from the analyses. The lower threshold was selected as a trade-off between the retention of valid observations and the exclusion of anticipatory responses [41]. The upper threshold was chosen based on prior work indicating this cut-off yields reliable estimates and maximises sensitivity [10], and was consistent with the range of reaction times observed in the KAAT pilot data [33]. Familiarisation trials (the first 15 trials of the study and the first 3 trials of each subsequent condition) will be removed from the analyses [18].

For the manikin task, reaction time was defined as the interval between stimulus onset and the key press. For the KAAT, reaction time was defined as the interval between peripheral target onset and movement initiation, which will be identified kinematically by first detecting the time point at which acceleration exceeds 1,000 mm/s² and then tracing backward to the last time point at which acceleration was above 200 mm/s². This procedure has been shown to provide a reliable and accurate estimate of movement initiation based on kinematic data [42].

Error Rate

Error rate was defined as the proportion of incorrect or time-out responses relative to the total number of trials, following the same exclusion criteria applied to reaction times. Higher values indicate poorer response accuracy.

Elapsed Time and Task Order

Elapsed time was defined as the duration between task onset (manikin or KAAT) and trial completion recorded for each trial. Task order (manikin first vs. KAAT first) will also be included as a categorical variable. These variables will account for potential temporal and order-related effects, including fatigue and learning.

Kinematic Variables

In addition to reaction time, three kinematic outcomes will be computed to capture complementary aspects of movement dynamics and approach-avoidance conflicts [43]: a derivative measure, a complexity measure, and a measure of deviation (Figure 6). When competing

approach and avoidance tendencies are concurrently activated during movement execution, participants may show altered movement dynamics as they resolve the conflict between task instructions and underlying tendencies. A summary of all dependent variables is provided in Supplemental Table 1.

The derivative measure, time to peak velocity, reflects reluctance to commit to action. It was defined as the time elapsed from movement onset (i.e., reaction time) to the time point of maximum movement velocity [44,45] on each correct trial. Longer times to peak velocity indicate delayed motor commitment.

The complexity measure, x-flips, reflects wavering during movement execution. It was defined as the number of directional changes along the x-axis [45,46] on each correct trial, which will be computed as the number of sign changes in velocity leading to a displacement greater than 1 mm. A higher number of x-flips indicates greater decision difficulty [47].

The deviation measure, opposite displacement, reflects response competition by capturing the extent to which the movement unfolds in the incorrect direction on each correct trial. Greater distance indicates greater response competition.

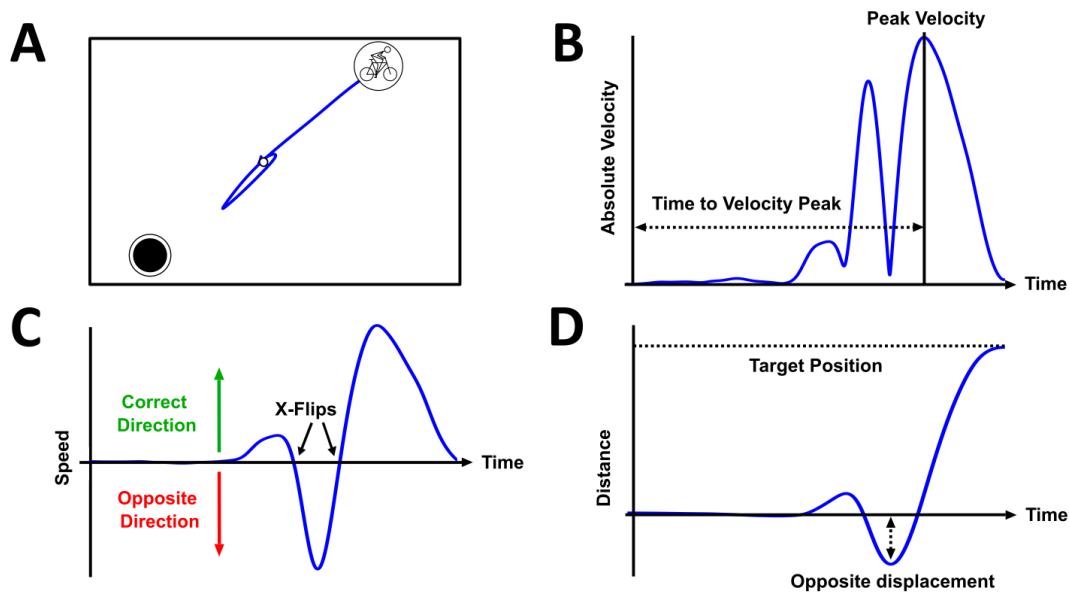


Figure 6. Kinematic Variables. (A) Example trial from the Kinarm approach-avoidance task (KAAT) with corresponding kinematic variables shown in panel (B) time to peak velocity, (C) number of x-flips, and (D) opposite displacement.

Raw and Relative Automatic Approach-Avoidance Bias

For construct and concurrent validity, the raw approach-avoidance bias will be estimated from the simple effect of Movement in the linear mixed-effects models, representing the Avoid – Approach contrast within each experimental stimulus category (physical activity or sedentary behaviour).

The relative bias will be estimated using the planned interaction contrasts of the Movement $\times$ Stimulus Type interaction. Specifically, for each experimental stimulus category, the contrast between avoidance and approach responses will be compared with the corresponding contrast for neutral stimuli (Equation 1), thereby accounting for a potential generic approach-avoidance tendency that may vary across participants [9,48]:

$$\beta_{\text{Movement} \times \text{Stimulus Type}} = (E[RT_{\text{Avoid Experimental}}] - E[RT_{\text{Approach Experimental}}]) - (E[RT_{\text{Avoid Neutral}}] - E[RT_{\text{Approach Neutral}}]) \quad (1)$$

where $E[RT]$ denotes the reaction time estimated by the model.

For the assessment of convergent and predictive validity, the relative bias towards physical activity will be calculated directly at the participant level using median reaction times (Equation 2). This is because mixed-effects models produce participant-specific estimates that are partially adjusted (or “shrunk”) towards the group average. While this shrinkage improves the stability of the estimates, it can attenuate (i.e., weaken) observed associations when these model-based estimates are subsequently used as predictors in separate analyses [49,50].

$$\text{Relative Bias}_{\text{Physical Activity}} = \left(\text{median } RT_{\text{Avoid Physical Activity}} - \text{median } RT_{\text{Approach Physical Activity}} \right) - \left(\text{median } RT_{\text{Avoid Neutral}} - \text{median } RT_{\text{Approach Neutral}} \right) \quad (2)$$

For conciseness, this section focuses on reaction time. However, raw and relative bias scores will be estimated separately for each kinematic variable. For reaction time, positive scores will indicate faster approach than avoidance responses to experimental stimuli, reflecting a tendency to approach rather than avoid. For kinematic variables, positive scores will indicate greater movement conflict during avoidance than approach trials, likewise reflecting a tendency to approach.

Reliability of Participant-Level Relative Bias

To evaluate the reliability of the relative approach-avoidance bias at the participant level, split-half reliability will be estimated using a resampling-based procedure combined with intraclass correlation coefficients (ICC).

For each participant and for each condition (physical activity and sedentary behaviour), trials will be randomly divided into two non-overlapping halves using stratified sampling to ensure that each half contains an equal number of observations from each combination of Movement (approach vs. avoidance) and Stimulus Type (experimental vs. neutral). Approach-avoidance relative bias score will be computed separately for each half of the data using linear mixed-effects models or, for median reaction times, using the same difference-of-differences formula described above, yielding two independent bias estimates per participant for each dependent variable.

To reduce dependency on any single arbitrary split and to obtain a more stable estimate of internal consistency, this procedure will be repeated 500 times using independent random stratified splits. For each iteration, agreement between the two bias estimates will be quantified using a two-way random-effects ICC with absolute agreement (ICC[2,1]) [51,52]. ICC values will then be averaged across iterations, and 95% confidence intervals will be reported to provide a robust estimate of reliability.

Because split-half reliability reflects the consistency of a half-length measure, coefficients will be adjusted using the Spearman-Brown prophecy formula to estimate reliability of the full-length task [53]. Higher ICC values will indicate greater within-participant stability and thus

greater reliability of the bias measure. Following conventional guidelines, ICC values < .40 will be interpreted as poor, .40–.59 as fair, .60–.74 as good, and $\geq .75$ as excellent reliability [52].

Device-Based Level of Physical Activity

Accelerometer Measurement

Device-based physical activity was assessed using a 3-axis ActiGraph wGT3X-BT accelerometer sampling at 30 Hz. The accelerometer was worn on the right hip using a belt clip and elastic waistband. Participants were instructed to wear the accelerometer for seven consecutive days during all waking hours, for at least 10 hours per day, except during aquatic activities (e.g., bathing, swimming). They also completed a daily sleep and wear log to support the identification of waking and non-wear periods. A monitoring week was considered valid if it included at least four valid days, including one weekend day [54–56]. Raw accelerometer files will be processed in R using the *GGIR* package [57,58]. Data will be auto-calibrated and summarised using the Euclidean Norm Minus One (ENMO) metric, expressed in milligravity units (mg). Waking wear time and non-wear periods will be identified using the sleep log in combination with the NotWorn and HDCZA algorithms, with non-wear detection based on a period of at least 60 consecutive minutes classified as non-wear by the GGIR algorithm. Physical activity intensity will be classified using adult hip-worn ENMO thresholds: inactivity, < 47.4 mg; light-intensity physical activity, 47.4 to < 69.1 mg; moderate-intensity physical activity, 69.1 to < 258.7 mg; and vigorous-intensity physical activity, ≥ 258.7 mg [59,60]. We anticipated that approximately 15% of participants would not provide valid accelerometer data [61]. Participants with insufficient valid accelerometer data or with data showing post-calibration error greater than 0.01 g (10 mg) will be excluded from device-based physical activity analyses [62], and the number and reasons for exclusions will be reported.

Total Physical Activity

Total physical activity will be computed to account for light-intensity physical activity, as this intensity has shown measurable health benefits [63] and has been shown to attenuate the association between sedentary time and mortality independently of moderate-to-vigorous physical activity [64]. Moreover, total physical activity volume across all intensities is more strongly associated with mortality reduction than moderate-to-vigorous physical activity alone [65], and has been adopted as a primary outcome in recent large accelerometer-based cohort studies [64,66].

Participants' time spent in each intensity category during the week will be averaged across all valid days to estimate average daily physical activity. Each intensity category will then be multiplied by the assigned metabolic equivalent of task (MET) to account for differences in energy expenditure across activity intensities: 2 METs for light physical activity, 4.5 METs for moderate physical activity, and 8 METs for vigorous physical activity [67]. The obtained weighted average values will finally be multiplied by seven to obtain a weekly total expressed in MET-min/week, reflecting weekly total physical activity volume adjusted for intensity. In addition, device-based moderate-to-vigorous physical activity in min/week will be calculated by averaging daily time spent in moderate and vigorous intensity activity and multiplying this value by seven.

Questionnaires

Demographic and Anthropometric Variables

Sex (male, female), age (years), height (cm), weight (kg), and handedness were self-reported.

Intention to Be Physically Active

Intention to be physically active was assessed with the item: “Over the next 7 days, I intend to do at least 150 minutes of moderate-intensity physical activity; or at least 75 minutes of vigorous intensity physical activity; or an equivalent combination of moderate- and vigorous-intensity physical activity”. Responses were provided on a 7-point Likert scale ranging from 1 (“strongly disagree”) to 7 (“strongly agree”) [9].

Explicit Affective Attitudes Towards Physical Activity

Explicit attitudes towards physical activity were assessed using two bipolar semantic differential items introduced by the stem: “For me, to participate in regular physical activity is ...” [9,68]. Responses were provided on a 7-point Likert scale for the adjective pairs “unpleasant” (1) to “pleasant” (7) and “unenjoyable” (1) to “enjoyable” (7). Scores were computed as the mean of the two items, with higher values indicating more positive affective attitudes towards physical activity.

Automaticity Towards Physical Activity

Automaticity towards physical activity was assessed using the automaticity subscale of the Self-Reported Habit Index and provided information on physical activity habit strength [69,70]. Participants first read the following stem “Overall, the decision to engage in physical activity is something...”. They then indicated their agreement with four statements: “I do automatically”, “I do without having to consciously remember”, “I do without thinking”, and “I start doing before I realize I’m doing it” ($\alpha = .94$). Responses were provided on a 7-point Likert scale ranging from 1 (“Strongly disagree”) to 7 (“Strongly agree”). Scores were computed as the mean of the four items, with higher scores indicating that engaging in physical activity is perceived as more automatic and habitual, whereas lower scores reflect more deliberate processes.

Valuation of Physical Effort

Valuation of physical effort was assessed using the Physical Effort Scale [71,72], an 8-item questionnaire with two dimensions: approach and avoidance of physical effort. Items were rated on a 5-point Likert scale ranging from 1 (“I completely disagree”) to 5 (“I completely agree”) and included statements such as “I usually like activities that require physical effort” and “I tend to avoid situations in which I have to exert physical effort”. Scores were computed as the mean of the eight items, with higher scores on the approach dimension reflecting greater positive valuation of physical effort, whereas higher scores on the avoidance dimension reflect stronger tendencies to avoid effort.

Descriptive Analyses

To characterise the sample, we will report means and standard deviations for age, height, weight, moderate-to-vigorous physical activity, total physical activity, intentions, explicit attitudes, automaticity, and the valuation of physical effort. Sex and handedness will be summarised using counts and percentages. Descriptive statistics will also be calculated for reaction times, number of errors, and kinematic variables, stratified by stimulus type and each movement type.

Statistical Analysis

Construct Validity (H1): Linear Mixed-Effects Models

To test H1, linear mixed-effects models will be fitted in the R software environment [73] using the *lme4* [74] and *lmerTest* packages [75]. The sub-hypotheses of H1 (H1a-H1t; Table 1) will be tested using the following model specification (Equation 3; R formula syntax):

$$\begin{aligned} \text{Dependent Variable} \sim & \text{Movement} * \text{Stimulus Type} * \text{Task} + \text{Age} + \text{Sex} + \\ & \text{Height} * \text{Weight} + \text{Handedness} + \text{Error Rate} + \text{Elapsed Time} + \text{Task Order} + \quad (3) \\ & (1 + \text{Task} + \text{Movement} * \text{Stimulus Type} | \text{Subject}) + (1 | \text{Pictogram}) \end{aligned}$$

Movement (approach and avoid) will be deviation-coded (-0.5 and $+0.5$, respectively) so that the Movement coefficient represents the Avoid – Approach contrast. Stimulus Type (neutral, physical activity and sedentary behaviour) will be entered as a three-level categorical predictor using treatment coding, with neutral stimuli as the reference category. Planned contrasts will therefore compare physical activity and sedentary behaviour stimuli separately against neutral stimuli. Task (manikin and KAAT) will be deviation-coded (-0.5 and $+0.5$, respectively) so that positive task-related interaction coefficients indicate larger approach-avoidance effects in the KAAT than in the manikin task. Continuous covariates will be standardised (z-scored) prior to analysis to facilitate model convergence and interpretation. Random effects will include intercepts for Subject and Pictogram. For participants, random slopes will be specified for Task and for the Movement $\times$ Stimulus Type effect, allowing individual differences in overall task effects and approach-avoidance patterns.

In the case of non-convergence or singular fit, model simplification will follow a predefined sequence. First, correlations between random slopes and intercepts will be removed. Second, random slopes associated with higher-order interaction terms will be sequentially removed. As a final step, covariates may be removed, while all primary fixed effects and their interactions will be retained. This predefined simplification strategy aims to balance model complexity with numerical stability and statistical power [76].

For the hypotheses testing internal structure focusing on the KAAT, raw approach-avoidance bias (H1a–h) will be tested using the simple effect of Movement (approach vs. avoidance) evaluated at the level of Task = KAAT, and the levels of Stimulus Type = physical activity or sedentary behaviour. KAAT relative approach-avoidance bias (H1i–p) will be tested using the Movement $\times$ Stimulus Type interaction, evaluated at Task = KAAT (see section *Raw and Relative Automatic Approach-Avoidance Bias*).

For the task-level differences in effect magnitude, difference in raw bias between tasks (H1q and H1r) will be tested using planned contrasts comparing the Movement effect between tasks at each experimental stimulus level. Difference in relative bias between tasks (H1s and H1t) will be tested using the Movement $\times$ Stimulus Type $\times$ Task interaction, which evaluates whether the neutral-corrected bias differs between the KAAT and the manikin task.

Models will be estimated using restricted maximum likelihood (REML). Statistical assumptions will be evaluated using residual diagnostics (Q-Q plots and residual-versus-fitted plots). If substantial deviations from normality or homoscedasticity are observed, sensitivity analyses will be conducted using log-transformed reaction times to assess the robustness of the findings. Statistical significance will be evaluated at $\alpha = .05$ using Satterthwaite-approximated degrees of freedom, and effect sizes with 95% confidence intervals will be reported.

Concurrent Validity (H2): Bivariate Correlations

To assess concurrent validity (H2), Pearson correlations will be computed between KAAT and manikin indices of raw approach-avoidance bias (H2a and H2b) and relative approach-avoidance bias (H2c and H2d), separately for physical activity and for sedentary stimuli. Correlation coefficients will be reported alongside 95% confidence intervals and corresponding p -values. The significance level will be set at $\alpha = .05$. Given prior evidence of limited convergence and substantial variability between approach-avoidance tasks [11,35], results will be interpreted with primary emphasis on effect size estimates and their precision rather than statistical significance alone.

Convergent Validity (H3): Multiple Linear Regressions

Convergent validity will be assessed by examining whether KAAT-derived relative approach-avoidance bias is associated with theoretically related behavioural and motivational indicators measured at the same time point (H3a–f), and whether these associations are stronger than those observed for the manikin task (H3g–l).

For H3a–f, separate multiple regressions will be estimated for each outcome variable: device-based total physical activity, device-based moderate-to-vigorous physical activity, explicit affective attitude towards physical activity, intention to be physically active, automaticity towards physical activity, and valuation of physical effort. Each model will include age, sex, height, weight, handedness, error rate, and elapsed time as covariates, with KAAT relative bias as the predictor of interest. The association between KAAT relative bias and each outcome will be evaluated using the standardised regression coefficient (β), its 95% confidence interval, and corresponding p -value.

For H3g–l, separate multiple regressions will include the same covariates, along with both KAAT relative bias and manikin relative bias. Including both biases in the same model allows estimation of their unique (partial) associations with each outcome while accounting for shared variance. Standardised regression coefficients and 95% confidence intervals will be reported for each task-specific bias. To formally test whether the KAAT relative bias is more strongly associated with each outcome than manikin bias, planned contrasts comparing regression coefficients ($\beta_{\text{KAAT}} - \beta_{\text{manikin}}$) will be conducted within each model. A significant positive contrast will indicate that the KAAT-derived bias explains more variance in the outcome than the manikin-derived bias.

All continuous covariates and predictors will be standardised prior to analysis. Multicollinearity will be assessed using variance inflation factors (VIF), with values above 5 considered indicative of potentially problematic collinearity. If detected, this will be transparently reported and considered in the interpretation of regression estimates rather than addressed through automatic variable exclusion [77].

To control for multiple comparisons within H3a–f and H3g–l [78], the six hypothesis tests in each set will be treated as a single family. P -values will be adjusted using the Benjamini-Hochberg False Discovery Rate procedure ($q = .05$) [79,80]. Device-based total physical activity will be specified a priori as the primary outcome within each family and will be evaluated without correction.

Predictive Validity (H4): Sequential Regressions and Model Comparison

Predictive validity will be evaluated by examining whether KAAT-derived measures explain variance in device-based total physical activity (H4a and H4b) and whether their predictive performance exceeds that of the manikin task (H4c).

For H4a and H4b, a multiple linear regression model will be estimated in three sequential steps. *In Step 1*, age, sex, height, weight, handedness, error rate, and elapsed time will be entered as covariates. *In Step 2*, KAAT relative bias for physical activity will be added. *In Step 3*: kinematic variables derived from the KAAT (time to peak velocity, number of x-flips, opposite displacement) will be added.

H4a will be evaluated by examining the change in explained variance (ΔR^2) when KAAT relative bias is added at Step 2. A statistically significant increase in ΔR^2 at this step will indicate that KAAT relative bias explains variance in device-based total physical activity beyond covariates. H4b will be evaluated by examining ΔR^2 when the kinematic variables are added at Step 3. A statistically significant increase in ΔR^2 at this step will indicate incremental predictive validity of the kinematic variables beyond relative bias. Changes in explained variance will be tested using F-tests for nested models. Standardised regression coefficients (β), 95% confidence intervals, and ΔR^2 values will be reported. Multicollinearity will be assessed as described for H3.

To test H4c, a separate regression including the same covariates as in Step 2 will be estimated, replacing the KAAT-derived relative bias with the manikin-derived relative bias. The predictive performance of the KAAT-based models (Step 2) and the manikin-based model will then be compared. Because these models are not nested, comparisons will rely on adjusted R^2 and information criteria (AIC and BIC) [81,82] rather than F-tests. Lower AIC and BIC values indicate better model fit. Differences in AIC (ΔAIC) will be interpreted using established guidelines [83], with values ≤ 2 indicating similar support for both models, values between 4 and 7 indicating less support for the model with the higher AIC, and values > 10 indicating no support for the model with the higher AIC. Differences in BIC (ΔBIC) will be interpreted using established guidelines [84], with values < 2 indicating negligible evidence, values between 2 and 6 indicating positive evidence, values between 6 and 10 indicating strong evidence, and values > 10 indicating very strong evidence against the model with the higher BIC. In addition, Akaike weights will be computed to quantify the relative support for each model, expressed as a probability-like measure of model plausibility [83,85].

Sensitivity Analyses

To evaluate the robustness of the main analyses, sensitivity analyses will be conducted for each reaction-time-based hypothesis using an upper reaction-time exclusion threshold of 3,000 ms instead of 1,500 ms. This threshold has previously been used to accommodate slower responses among older participants [9], and may minimise the risk of influencing results through the exclusion of potentially valid observations [41]. These analyses will assess whether the estimated approach-avoidance bias scores remain robust across alternative reaction-time processing thresholds.

Data & Code Accessibility

Supplemental material, data, R scripts, and material used for this study are publicly available on GitHub: https://github.com/Boisgontier-Lab/RR_Validation_KAAT [86].

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Ethical Statement

The study was approved by University of Ottawa's Research Ethics Boards (H-01-25-11075) and informed consent was collected in accordance with the Declaration of Helsinki.

Use of Artificial Intelligence (AI) and AI-assisted technologies

ChatGPT (OpenAI) and DeepL were used to refine the language, improve readability, and correct grammar and spelling errors of this manuscript.

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Competing Interests

Matthieu P. Boisgontier is the founder, representative, and publication director of Peer Community In Health & Movement Sciences (PCI HMS), and serves as a Recommender for PCI Neuroscience and PCI Registered Reports. François Jabouille is a member of the Managing Board of PCI HMS. Kayne Park is a Recommender for PCI HMS.

Authors' Contributions

Based on the Contributor Roles Taxonomy (CRediT) [87] individual author contributions to this work will be as follows: François Jabouille: Conceptualisation, Methodology, Investigation, Data Curation, Formal Analysis, Visualisation, Writing – Original Draft, Writing – Review and Editing, Supervision; Timothée Dumas: Data Curation, Visualisation, Writing – Review and Editing; Jérémy Fortier: Investigation; Michaël Patrick Robichaud: Investigation; Brian Nhat Thien Nguyen: Investigation; Frédérik Jolicoeur: Investigation; Silvio Maltagliati: Writing – Review and Editing; Boris Cheval: Writing – Review and Editing; Kayne Park: Methodology, Data Curation, Visualisation, Writing – Review and Editing; Matthieu Boisgontier: Conceptualisation, Methodology, Formal Analysis, Visualisation, Writing – Original Draft, Writing – Review and Editing, Supervision, Project Administration, Funding Acquisition.

REFERENCES

1. Cheval B, Radel R, Neva JL, Boyd LA, Swinnen SP, Sander D, Boisgontier MP. 2018 Behavioral and neural evidence of the rewarding value of exercise behaviors: a systematic review. *Sports Med.* 48, 1389–1404. ([doi:10.1007/s40279-018-0898-0](https://doi.org/10.1007/s40279-018-0898-0))
2. Brand R, Ekkekakis P. 2018 Affective–Reflective Theory of physical inactivity and exercise: Foundations and preliminary evidence. *Ger. J. Exerc. Sport Res.* 48, 48–58. ([doi:10.1007/s12662-017-0477-9](https://doi.org/10.1007/s12662-017-0477-9))
3. Conroy DE, Berry TR. 2017 Automatic affective evaluations of physical activity. *Exerc. Sport Sci. Rev.* 45, 230–237. ([doi:10.1249/JES.0000000000000120](https://doi.org/10.1249/JES.0000000000000120))
4. Cheval B, Boisgontier MP. 2021 The theory of effort minimization in physical activity. *Exerc. Sport Sci. Rev.* 49, 168–178. ([doi:10.1249/JES.0000000000000252](https://doi.org/10.1249/JES.0000000000000252))
5. Strack F, Deutsch R. 2004 Reflective and impulsive determinants of social behavior. *Pers. Soc. Psychol. Rev.* 8, 220–247. ([doi:10.1207/s15327957pspr0803_1](https://doi.org/10.1207/s15327957pspr0803_1))
6. Hofmann W, Friese M, Wiers RW. 2008 Impulsive versus reflective influences on health behavior: a theoretical framework and empirical review. *Health Psychol. Rev.* 2, 111–137. ([doi:10.1080/17437190802617668](https://doi.org/10.1080/17437190802617668))
7. Chen M, Bargh JA. 1999 Consequences of automatic evaluation: immediate behavioral predispositions to approach or avoid the stimulus. *Pers. Soc. Psychol. Bull.* 25, 215–224. ([doi:10.1177/0146167299025002007](https://doi.org/10.1177/0146167299025002007))
8. Rinck M, Becker ES. 2007 Approach and avoidance in fear of spiders. *J. Behav. Ther. Exp. Psychiatry* 38, 105–120. ([doi:10.1016/j.jbtep.2006.10.001](https://doi.org/10.1016/j.jbtep.2006.10.001))
9. Farajzadeh A, Goubran M, Beehler A, Cherkaoui N, Morrison P, De Chanaleilles M, Maltagliati S, Boris C, Miller MW, Sheehy L, et al. 2023 Automatic approach-avoidance tendency toward physical activity, sedentary, and neutral stimuli as a function of age, explicit affective attitude, and intention to be active. *Peer Community J.* 3, e21. ([doi:10.24072/pcjournal.246](https://doi.org/10.24072/pcjournal.246))
10. Krieglmeier R, Deutsch R. 2010 Comparing measures of approach–avoidance behaviour: the manikin task vs. two versions of the joystick task. *Cogn. Emot.* 24, 810–828. ([doi:10.1080/02699930903047298](https://doi.org/10.1080/02699930903047298))
11. Wittekind CE, Blechert J, Schiebel T, Lender A, Kahveci S, Kühn S. 2021 Comparison of different response devices to assess behavioral tendencies towards chocolate in the approach-avoidance task. *Appetite* 165, 105294. ([doi:10.1016/j.appet.2021.105294](https://doi.org/10.1016/j.appet.2021.105294))
12. Braunstein LM, Kuerbis A, Ochsner K, Morgenstern J. 2016 Implicit alcohol approach and avoidance tendencies predict future drinking in problem drinkers. *Alcohol Clin. Exp. Res.* 40, 1945–1952. ([doi:10.1111/acer.13151](https://doi.org/10.1111/acer.13151))
13. Evans TC, Taylor CT, Britton JC. 2021 Characterizing the time course of automatic action tendencies to affective facial expressions and its dysregulation in social anxiety disorder. *J. Anxiety Disord.* 78, 102363. ([doi:10.1016/j.janxdis.2021.102363](https://doi.org/10.1016/j.janxdis.2021.102363))
14. Deuter CE, Smit J, Kaczmarczyk M, Wingenfeld K, Otte C, Kuehl LK. 2021 Approach-avoidance tendencies in depression and childhood trauma: no effect of noradrenergic stimulation. *Compr. Psychoneuroendocrinol.* 8, 100077. ([doi:10.1016/j.cpne.2021.100077](https://doi.org/10.1016/j.cpne.2021.100077))

15. Cheval B, Sarrazin P, Pelletier L. 2014 Impulsive approach tendencies towards physical activity and sedentary behaviors, but not reflective intentions, prospectively predict non-exercise activity thermogenesis. *PLoS One* 9, e115238. ([doi:10.1371/journal.pone.0115238](https://doi.org/10.1371/journal.pone.0115238))
16. Cheval B, Tipura E, Burra N, Frossard J, Chanal J, Orsholits D, Radel R, Boisgontier MP. 2018 Avoiding sedentary behaviors requires more cortical resources than avoiding physical activity: An EEG study. *Neuropsychologia* 119, 68–80. ([doi:10.1016/j.neuropsychologia.2018.07.029](https://doi.org/10.1016/j.neuropsychologia.2018.07.029))
17. Cheval B, Sarrazin P, Isoard-Gautheur S, Radel R, Friese M. 2015 Reflective and impulsive processes explain (in)effectiveness of messages promoting physical activity: a randomized controlled trial. *Health Psychol.* 34, 10–19. ([doi:10.1037/hea0000102](https://doi.org/10.1037/hea0000102))
18. Farajzadeh A, Jabouille F, Benoit N, Bezeau O, Bourgie T, Gerro B, Ouimet J, Boisgontier MP. 2024 Apathy, intentions, explicit attitudes, and approach-avoidance tendencies in physical activity behavior. *Commun. Kinesiol.* 1, 68. ([doi:10.51224/cik.2024.68](https://doi.org/10.51224/cik.2024.68))
19. Cheval B, Sarrazin P, Isoard-Gautheur S, Radel R, Friese M. 2016 How impulsivity shapes the interplay of impulsive and reflective processes involved in objective physical activity. *Pers. Individ. Dif.* 96, 132–137. ([doi:10.1016/j.paid.2016.02.067](https://doi.org/10.1016/j.paid.2016.02.067))
20. Daou M, Lohse K, Miller M. 2017 To take the stairs or not to take the stairs? Employing the reflective–impulsive model to predict spontaneous physical activity. *Sports* 5, 75. ([doi:10.3390/sports5040075](https://doi.org/10.3390/sports5040075))
21. Oliver S, Kemps E. 2018 Motivational and implicit processes contribute to incidental physical activity. *Br. J. Health Psychol.* 23, 820–842. ([doi:10.1111/bjhp.12318](https://doi.org/10.1111/bjhp.12318))
22. Hannan TE, Moffitt RL, Neumann DL, Kemps E. 2019 Implicit approach–avoidance associations predict leisure-time exercise independently of explicit exercise motivation. *Sport Exerc. Perform. Psychol.* 8, 210–222. ([doi:10.1037/spy0000145](https://doi.org/10.1037/spy0000145))
23. Zenko Z, Ekkekakis P. 2019 Internal consistency and validity of measures of automatic exercise associations. *Psychol. Sport Exerc.* 43, 4–15. ([doi:10.1016/j.psychsport.2018.12.005](https://doi.org/10.1016/j.psychsport.2018.12.005))
24. Berry TR. 2020 You do (not?) have to go to the gym: Effects of negations in exercise messages. *Sport Exerc. Perform. Psychol.* 9, 437–449. ([doi:10.1037/spy0000185](https://doi.org/10.1037/spy0000185))
25. Moffitt RL, Kemps E, Hannan TE, Neumann DL, Stopar SP, Anderson CJ. 2020 Implicit approach biases for physically active lifestyle cues. *Int. J. Sport Exerc. Psychol.* 18, 833–849. ([doi:10.1080/1612197X.2019.1581829](https://doi.org/10.1080/1612197X.2019.1581829))
26. Wang Y-B, Sun X-T, Mao Z-X. 2023 Is approaching sedentary behaviour or physical activity a reward? An EEG study. *Behav. Brain Res.* 445, 114364. ([doi:10.1016/j.bbr.2023.114364](https://doi.org/10.1016/j.bbr.2023.114364))
27. Fessler L, Tessitore E, Craviari C, Sarrazin P, Meyer P, Luthy C, Hanna-Deschamps E, Cheval B. 2024 Motivational and emotional correlates of physical activity and sedentary behavior after cardiac rehabilitation: an observational study. *BMC Sports Sci. Med. Rehabil.* 16, 209. ([doi:10.1186/s13102-024-00997-0](https://doi.org/10.1186/s13102-024-00997-0))
28. Goubiran M, Zammar C, Tellez Alvarez S, Héran É, Proulx S, Bilodeau M, Boisgontier MP. 2025 Approach-avoidance tendencies moderate the relationship between fear of movement and physical activity in osteoarthritis. *Physiother. Can.* e20250011. ([doi:10.3138/ptc-2025-0011](https://doi.org/10.3138/ptc-2025-0011))

29. Wang C, Zhang G. 2025 Examination of the consistency effect between physical activity levels and approach-avoidance behaviors. *Psychol. Sport Exerc.* 79, 102825. ([doi:10.1016/j.psychsport.2025.102825](https://doi.org/10.1016/j.psychsport.2025.102825))
30. Preis MA, Zellerhoff M, Brockmeyer T. 2021 Approach bias modification training to increase physical activity: a pilot randomized controlled trial in healthy volunteers. *J. Health Psychol.* 26, 2470–2486. ([doi:10.1177/1359105320913936](https://doi.org/10.1177/1359105320913936))
31. Locke SR, Berry TR. 2021 Examining the relationship between exercise-related cognitive errors, exercise schema, and implicit associations. *J. Sport Exerc. Psychol.* 43, 345–352. ([doi:10.1123/jsep.2021-0031](https://doi.org/10.1123/jsep.2021-0031))
32. Quossi A, Fessler L, Maltagliati S, Gardner B, Miller MW, Sander D, Rebar AL, Szabo A, Cheval B. 2025 Can automatic reactions mirror exercise dependence? *Int. J. Sport Exerc. Psychol.* 23, 1207–1233. ([doi:10.1080/1612197X.2024.2387698](https://doi.org/10.1080/1612197X.2024.2387698))
33. Park K, Boisgontier MP. 2025 A novel robotic reaching task to advance the assessment of approach-avoidance tendencies through kinematic analysis. *Peer Community J.* 5, e25. ([doi:10.24072/pcjournal.530](https://doi.org/10.24072/pcjournal.530))
34. Indu P, Vidhukumar K, Chacko D, Menon V, Grover S, Gupta S. 2025 Criterion validity, construct validity, and factor analysis: an introductory overview. *Indian J. Psychiatry* 67, 916–921. ([doi:10.4103/indianjpsychiatry_911_25](https://doi.org/10.4103/indianjpsychiatry_911_25))
35. Fricke K, Alexander N, Jacobsen T, Vogel S. 2023 Comparison of two reaction-time-based and one foraging-based behavioral approach-avoidance tasks in relation to interindividual differences and their reliability. *Sci. Rep.* 13, 22376. ([doi:10.1038/s41598-023-49864-x](https://doi.org/10.1038/s41598-023-49864-x))
36. Green P, MacLeod CJ. 2016 SIMR: an R package for power analysis of generalized linear mixed models by simulation. *Methods Ecol. Evol.* 7, 493–498. ([doi:10.1111/2041-210X.12504](https://doi.org/10.1111/2041-210X.12504))
37. Cheval B, Tipura E, Burra N, Frossard J, Chanal J, Orsholits D, Radel R, Boisgontier MP. 2018 Avoiding sedentary behaviors requires more cortical resources than avoiding physical activity. Zenodo. 1169140. ([doi:10.5281/zenodo.1169140](https://doi.org/10.5281/zenodo.1169140))
38. Morris TP, White IR, Crowther MJ. 2019 Using simulation studies to evaluate statistical methods. *Stat. Med.* 38, 2074–2102. ([doi:10.1002/sim.8086](https://doi.org/10.1002/sim.8086))
39. Lakens D. 2022 Sample size justification. *Collabra Psychol.* 8, 33267. ([doi:10.1525/collabra.33267](https://doi.org/10.1525/collabra.33267))
40. Jabouille F, Park K, Boisgontier M. 2026 Manikin and KAAT demonstration videos. GitHub. https://github.com/Boisgontier-Lab/RR_Validation_KAAT/tree/main/Manikin_and_KAAT_demonstration_videos
41. Miller J. 2023 Outlier exclusion procedures for reaction time analysis: the cures are generally worse than the disease. *J. Exp. Psychol. Gen.* 152, 3189–3217. ([doi:10.1037/xge0001450](https://doi.org/10.1037/xge0001450))
42. Blinch J, Trovinger C, DeWinne CR, De Cellio Martins G, Ifediora CN, Nourollahimoghdam M, Harry JR, Palmer TB. 2023 Tradeoffs of estimating reaction time with absolute and relative thresholds. *Behav. Res. Methods* 56, 4695–4715. ([doi:10.3758/s13428-023-02211-4](https://doi.org/10.3758/s13428-023-02211-4))
43. Garcia-Guerrero S, O’Hora D, Zgonnikov A, Scherbaum S. 2023 The action dynamics of approach-avoidance conflict during decision-making. *Q. J. Exp. Psychol. (Hove)* 76, 160–179. ([doi:10.1177/17470218221087625](https://doi.org/10.1177/17470218221087625))

44. Grenke O, Scherbaum S, Schoemann M. 2025 Exploring the synthesis of mouse cursor tracking and drift diffusion modeling in a perceptual decision-making task. *Behav. Res. Methods* 57, 297. ([doi:10.3758/s13428-025-02805-0](https://doi.org/10.3758/s13428-025-02805-0))
45. Wirth R, Foerster A, Kunde W, Pfister R. 2020 Design choices: empirical recommendations for designing two-dimensional finger-tracking experiments. *Behav. Res. Methods* 52, 2394–2416. ([doi:10.3758/s13428-020-01409-0](https://doi.org/10.3758/s13428-020-01409-0))
46. Dale R, Roche J, Snyder K, McCall R. 2008 Exploring action dynamics as an index of paired-associate learning. *PLoS One* 3, e1728. ([doi:10.1371/journal.pone.0001728](https://doi.org/10.1371/journal.pone.0001728))
47. Cheng J, González-Vallejo C. 2018 Unpacking decision difficulty: testing action dynamics in Intertemporal, gamble, and consumer choices. *Acta Psychol. (Amst)* 190, 199–216. ([doi:10.1016/j.actpsy.2018.08.002](https://doi.org/10.1016/j.actpsy.2018.08.002))
48. Windsor TD, Pearson EL, Butterworth P. 2012 Age group differences and longitudinal changes in approach–avoidance sensitivity: findings from an 8-year longitudinal study. *J. Res. Pers.* 46, 646–654. ([doi:10.1016/j.jrp.2012.07.002](https://doi.org/10.1016/j.jrp.2012.07.002))
49. Kliegl R, Wei P, Dambacher M, Yan M, Zhou X. 2010 Experimental effects and individual differences in linear mixed models: estimating the relationship between spatial, object, and attraction effects in visual attention. *Front. Psychol.* 1, 238. ([doi:10.3389/fpsyg.2010.00238](https://doi.org/10.3389/fpsyg.2010.00238))
50. Liu S, Kuppens P, Bringmann L. 2021 On the use of empirical Bayes estimates as measures of individual traits. *Assessment* 28, 845–857. ([doi:10.1177/1073191119885019](https://doi.org/10.1177/1073191119885019))
51. Shrout PE, Fleiss JL. 1979 Intraclass correlations: uses in assessing rater reliability. *Psychol. Bull.* 86, 420–428. ([doi:10.1037/0033-2909.86.2.420](https://doi.org/10.1037/0033-2909.86.2.420))
52. Koo TK, Li MY. 2016 A guideline of selecting and reporting intraclass correlation coefficients for reliability research. *J. Chiropr. Med.* 15, 155–163. ([doi:10.1016/j.jcm.2016.02.012](https://doi.org/10.1016/j.jcm.2016.02.012))
53. de Vet HCW, Mokkink LB, Mosmuller DG, Terwee CB. 2017 Spearman–Brown prophecy formula and Cronbach’s alpha: different faces of reliability and opportunities for new applications. *J. Clin. Epidemiol.* 85, 45–49. ([doi:10.1016/j.jclinepi.2017.01.013](https://doi.org/10.1016/j.jclinepi.2017.01.013))
54. Migueles JH, Cadenas-Sanchez C, Ekelund U, Delisle Nyström C, Mora-Gonzalez J, Löf M, Labayen I, Ruiz JR, Ortega FB. 2017 Accelerometer data collection and processing criteria to assess physical activity and other outcomes: a systematic review and practical considerations. *Sports Med.* 47, 1821–1845. ([doi:10.1007/s40279-017-0716-0](https://doi.org/10.1007/s40279-017-0716-0))
55. Trost SG, McIver KL, Pate RR. 2005 Conducting accelerometer-based activity assessments in field-based research. *Med. Sci. Sports Exerc.* 37, S531–S543. ([doi:10.1249/01.mss.0000185657.86065.98](https://doi.org/10.1249/01.mss.0000185657.86065.98))
56. Suorsa K, Leskinen T, Rovio S, Niinikoski H, Pentti J, Nevalainen J, Heinonen OJ, Lagström H, Jula A, Viikari J, et al. 2023 Weekday and weekend physical activity patterns and their correlates among young adults. *Scand. J. Med. Sci. Sports* 33, 2573–2584. ([doi:10.1111/sms.14475](https://doi.org/10.1111/sms.14475))
57. Migueles JH, Rowlands AV, Huber F, Sabia S, Van Hees VT. 2019 GGIR: a research community–driven open source R package for generating physical activity and sleep outcomes from multi-day raw accelerometer data. *J. Meas. Phys. Behav.* 2, 188–196. ([doi:10.1123/jmpb.2018-0063](https://doi.org/10.1123/jmpb.2018-0063))

58. Van Hees VT, Migueles JH, Sabia S, Patterson MR, Fang Z, Heywood J, Capdevila Pujol J, Kushleyeva L, Chen M, Yerramalla M, et al. 2026 GGIR: raw accelerometer data analysis. R package version 3.3-7. Zenodo. ([doi:10.5281/zenodo.1051064](https://doi.org/10.5281/zenodo.1051064))
59. Hildebrand M, Hansen BH, Van Hees VT, Ekelund U. 2017 Evaluation of raw acceleration sedentary thresholds in children and adults. *Scand. J. Med. Sci. Sports* 27, 1814–1823. ([doi:10.1111/sms.12795](https://doi.org/10.1111/sms.12795))
60. Hildebrand M, Van Hees VT, Hansen BH, Ekelund U. 2014 Age group comparability of raw accelerometer output from wrist- and hip-worn monitors. *Med. Sci. Sports Exerc.* 46, 1816–1824. ([doi:10.1249/MSS.0000000000000289](https://doi.org/10.1249/MSS.0000000000000289))
61. Colley R, Connor Gorber S, Tremblay MS. 2010 Quality control and data reduction procedures for accelerometry-derived measures of physical activity. *Health Rep.* 21, 63–69.
62. Van Hees VT, Fang Z, Langford J, Assah F, Mohammad A, da Silva ICM, Trenell MI, White T, Wareham NJ, Brage S. 2014 Autocalibration of accelerometer data for free-living physical activity assessment using local gravity and temperature: an evaluation on four continents. *J. Appl. Physiol.* 117, 738–744. ([doi:10.1152/jappphysiol.00421.2014](https://doi.org/10.1152/jappphysiol.00421.2014))
63. Amagasa S, Machida M, Fukushima N, Kikuchi H, Takamiya T, Odagiri Y, Inoue S. 2018 Is objectively measured light-intensity physical activity associated with health outcomes after adjustment for moderate-to-vigorous physical activity in adults? A systematic review. *Int. J. Behav. Nutr. Phys. Act.* 15, 65. ([doi:10.1186/s12966-018-0695-z](https://doi.org/10.1186/s12966-018-0695-z))
64. Rezende LFM, Ahmadi M, Ferrari G, Del Pozo Cruz B, Lee I-M, Ekelund U, Stamatakis E. 2024 Device-measured sedentary time and intensity-specific physical activity in relation to all-cause and cardiovascular disease mortality: the UK Biobank cohort study. *Int. J. Behav. Nutr. Phys. Act.* 21, 68. ([doi:10.1186/s12966-024-01615-5](https://doi.org/10.1186/s12966-024-01615-5))
65. Tarp J, Dalene KE, Fagerland MW, Steene-Johannesen J, Hansen BH, Anderssen SA, Hagströmer M, Dohrn IM, Dempsey PC, Wijndaele K, et al. 2024 Physical activity volume, intensity, and mortality: harmonized meta-analysis of prospective cohort studies. *Am. J. Prev. Med.* 67, 887–896. ([doi:10.1016/j.amepre.2024.07.022](https://doi.org/10.1016/j.amepre.2024.07.022))
66. Watts EL et al. 2023 Association of accelerometer-measured physical activity level with risks of hospitalization for 25 common health conditions in UK adults. *JAMA Netw. Open* 6, e2256186. ([doi:10.1001/jamanetworkopen.2022.56186](https://doi.org/10.1001/jamanetworkopen.2022.56186))
67. Colpitts BH, Mayo A, Bélanger M, Bouchard DR, Boudreau J, Rioux BV, Sénéchal M. 2021 Exploring the association of total physical activity energy expenditure and diabetes mellitus in adults: a cross-sectional analysis of the Canadian Health Measures Survey. *Metab. Syndr. Relat. Disord.* 19, 556–561. ([doi:10.1089/met.2021.0024](https://doi.org/10.1089/met.2021.0024))
68. Hoyt AL, Rhodes RE, Hausenblas HA, Giacobbi PR. 2009 Integrating five-factor model facet-level traits with the theory of planned behavior and exercise. *Psychol. Sport Exerc.* 10, 565–572. ([doi:10.1016/j.psychsport.2009.02.008](https://doi.org/10.1016/j.psychsport.2009.02.008))
69. Verplanken B, Orbell S. 2003 Reflections on past behavior: a Self-Report Index of Habit Strength. *J. Appl. Soc. Psychol.* 33, 1313–1330. ([doi:10.1111/j.1559-1816.2003.tb01951.x](https://doi.org/10.1111/j.1559-1816.2003.tb01951.x))
70. Gardner B, Abraham C, Lally P, De Bruijn GJ. 2012 Towards parsimony in habit measurement: Testing the convergent and predictive validity of an automaticity subscale of the Self-Report Habit Index. *Int. J. Behav. Nutr. Phys. Act.* 9, 102. ([doi:10.1186/1479-5868-9-102](https://doi.org/10.1186/1479-5868-9-102))

71. St-Denis B, Beaudry S, Boisgontier MP, Cheval B, Maltagliati S. 2024 Validation de la version francophone de l'Échelle d'Effort Physique. *Mov. Sport Sci.* 126, 33–45. ([doi:10.1051/sm/2024019](https://doi.org/10.1051/sm/2024019))
72. Cheval B, Maltagliati S, Courvoisier DS, Marcora S, Boisgontier MP. 2024 Development and validation of the physical effort scale (PES). *Psychol. Sport Exerc.* 72, 102607. ([doi:10.1016/j.psychsport.2024.102607](https://doi.org/10.1016/j.psychsport.2024.102607))
73. R Core Team. 2024 R: A language and environment for statistical computing, version 4.4.2. Vienna, Austria: R Foundation for Statistical Computing. <https://www.R-project.org/>
74. Bates D, Mächler M, Bolker B, Walker S. 2015 Fitting linear mixed-effects models using lme4. *J. Stat. Softw.* 67, 1–48. ([doi:10.18637/jss.v067.i01](https://doi.org/10.18637/jss.v067.i01))
75. Kuznetsova A, Brockhoff PB, Christensen RHB. 2017 lmerTest package: tests in linear mixed effects models. *J. Stat. Softw.* 82, 1–26. ([doi:10.18637/jss.v082.i13](https://doi.org/10.18637/jss.v082.i13))
76. Matuschek H, Kliegl R, Vasishth S, Baayen H, Bates D. 2017 Balancing Type I error and power in linear mixed models. *J. Mem. Lang.* 94, 305–315. ([doi:10.1016/j.jml.2017.01.001](https://doi.org/10.1016/j.jml.2017.01.001))
77. O'Brien RM. 2007 A caution regarding rules of thumb for variance inflation factors. *Qual. Quant.* 41, 673–690. ([doi:10.1007/s11135-006-9018-6](https://doi.org/10.1007/s11135-006-9018-6))
78. Bender R, Lange S. 2001 Adjusting for multiple testing—when and how? *J. Clin. Epidemiol.* 54, 343–349. ([doi:10.1016/S0895-4356\(00\)00314-0](https://doi.org/10.1016/S0895-4356(00)00314-0))
79. Glickman ME, Rao SR, Schultz MR. 2014 False discovery rate control is a recommended alternative to Bonferroni-type adjustments in health studies. *J. Clin. Epidemiol.* 67, 850–857. ([doi:10.1016/j.jclinepi.2014.03.012](https://doi.org/10.1016/j.jclinepi.2014.03.012))
80. Benjamini Y, Hochberg Y. 1995 Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J. R. Stat. Soc. Ser. B Methodol.* 57, 289–300. ([doi:10.1111/j.2517-6161.1995.tb02031.x](https://doi.org/10.1111/j.2517-6161.1995.tb02031.x))
81. Akaike H. 1974 A new look at the statistical model identification. *IEEE Trans. Automat. Contr.* 19, 716–723. ([doi:10.1109/TAC.1974.1100705](https://doi.org/10.1109/TAC.1974.1100705))
82. Schwarz G. 1978 Estimating the dimension of a model. *Ann. Statist.* 6, 461–464. ([doi:10.1214/aos/1176344136](https://doi.org/10.1214/aos/1176344136))
83. Burnham KP, Anderson DR. 2002 *Model selection and multimodel inference: a practical information-theoretic approach*. 2nd ed. New York: Springer.
84. Kass RE, Raftery AE. 1995 Bayes Factors. *J. Am. Stat. Assoc.* 90, 773–795. ([doi:10.1080/01621459.1995.10476572](https://doi.org/10.1080/01621459.1995.10476572))
85. Wagenmakers E-J, Farrell S. 2004 AIC model selection using Akaike weights. *Psychon. Bull. Rev.* 11, 192–196. ([doi:10.3758/BF03206482](https://doi.org/10.3758/BF03206482))
86. Jabouille F, Park K, Boisgontier M. 2026 RR_Validation_KAAT. GitHub. https://github.com/Boisgontier-Lab/RR_Validation_KAAT
87. Allen L, O'Connell A, Kiermer V. 2019 How can we ensure visibility and diversity in research contributions? How the Contributor Role Taxonomy (CRediT) is helping the shift from authorship to contributorship. *Learn. Publ.* 32, 71–74. ([doi:10.1002/leap.1210](https://doi.org/10.1002/leap.1210))

TABLES

Table 2. Study design

Question	Hypotheses	Sampling plan	Analysis plan	Rationale for deciding the sensitivity of the test for confirming or disconfirming the hypothesis	Interpretation given different outcomes
What is the construct validity of the KAAT?	Validity was assessed based on raw bias (difference between avoidance and approach responses in experimental conditions) and relative bias (same difference corrected for the corresponding neutral-stimulus conditions) computed from all dependent variables (reaction time, time to peak velocity, x-flips, and opposite displacement). Hypotheses: - KAAT-derived raw biases would significantly differ from zero for physical activity and sedentary stimuli (H1a-h). - KAAT-derived relative biases would significantly differ from zero for physical activity and sedentary stimuli (H1i-p). - Raw bias would be larger when assessed	The following linear mixed-effects model was fitted separately for physical activity and sedentary stimuli: Reaction Time ~ Movement × Stimulus Type + (1 Movement Subject) + (1 Pictogram), with Movement = approach or avoidance response, and Stimulus Type = physical activity and neutral pictograms, or sedentary and neutral pictograms, depending on the model. Fixed-effect estimates, random-effect variance components (VarCorr), and residual variance (σ^2) from these pilot models were then used to generate simulated datasets reproducing the planned balanced experimental design for each experimental stimulus category. Statistical power for the Movement × Stimulus Type interaction was estimated using 1000	The sub-hypotheses of H1 (H1a-H1t; Table 1) will be tested using the following model specification: Dependent Variable ~ Movement * Stimulus Type * Task + Age + Sex + Height*Weight + Handedness + Error Rate + Elapsed Time + Task Order + (1 + Task + Movement * Stimulus Type Subject) + (1 Pictogram) Movement (approach and avoid) and Task (manikin and KAAT) will be deviation-coded (-0.5 and +0.5, respectively) so that the Movement coefficient represents the Avoid – Approach contrast. Stimulus Type (neutral, physical activity and sedentary behaviour) will be entered as a three-level categorical predictor using treatment coding, with neutral stimuli as the reference category. Planned contrasts will compare physical activity and sedentary behaviour stimuli separately against neutral stimuli. Random effects will include intercepts for Subject and Pictogram. For participants, random slopes will be specified for Task and for the Movement × Stimulus Type effect, allowing individual differences in overall task effects and approach-avoidance patterns. - KAAT raw approach-avoidance bias (H1a-h) will be tested using the simple effect of Movement (approach vs. avoidance) evaluated at the level of Task	- The final sample (N = 96) exceeds the minimum required sample size and is therefore expected to provide adequate sensitivity to detect effects of the two-way interaction. - No simulation was conducted for the three-way interaction comparing tasks (H1q-t) because the KAAT is new and reliable estimates for between-task differences are unavailable. Conclusions regarding task differences will therefore rely particularly on effect-size estimates and 95% confidence intervals.	For all the hypotheses, non-significant effects will be interpreted as insufficient evidence to support the corresponding hypothesis. For each experimental stimulus category: - A significant Simple effect of Movement will be interpreted as evidence of a raw approach-avoidance effect. - A significant Movement × Task interaction will suggest that the magnitude of the raw approach-avoidance bias depends on the type of task used. Specifically, a positive interaction effect will be interpreted as evidence that the KAAT elicits larger approach-avoidance effects than the manikin task. A negative interaction effect will be interpreted as the opposite. - A significant Movement × Stimulus Type interaction will be interpreted as evidence of a relative approach-avoidance effect. - A significant Movement × Stimulus Type × Task interaction will be interpreted as evidence that the magnitude of the relative approach-avoidance bias depends on the type of task used. Specifically, a positive interaction effect will be interpreted as evidence that the KAAT elicits larger approach-avoidance effects, whereas a negative interaction effect will be interpreted as the opposite.

	with the KAAT than with the manikin task for both physical activity (H1q) and sedentary stimuli (H1r). - Relative bias would be larger when assessed with the KAAT than with the manikin task for both physical activity (H1s) and sedentary stimuli (H1t).	Monte-Carlo simulations ($\alpha = .05$; t-test with Satterthwaite-approximated degrees of freedom on the fixed effect). Simulation-based power curves indicated that achieving 90% power required 32 participants for physical activity stimuli and 45 participants for sedentary stimuli (Figure 1). We therefore set 45 participants as the minimum required sample size.	= KAAT, and the levels of Stimulus Type = physical activity or sedentary behaviour. - KAAT relative approach-avoidance bias (H1i-p) will be tested using the Movement $\times$ Stimulus Type interaction, evaluated at Task = KAAT. - Difference in raw bias between tasks (H1q and H1r) will be tested using the Movement $\times$ Task interaction, evaluated separately at the physical-activity and sedentary-behaviour levels of experimental stimuli. - Difference in relative bias between tasks (H1s and H1t) will be tested using the Movement $\times$ Stimulus Type $\times$ Task interaction, which evaluates whether the neutral-corrected bias differs between the KAAT and the manikin task.		
What is the concurrent validity of the KAAT?	Hypotheses: - raw bias would show small positive correlations between tasks for both physical activity (H2a) and sedentary stimuli (H2b) - relative bias would show small positive correlations between tasks for both physical activity (H2c) and sedentary stimuli (H2d).		Pearson correlations will be computed between KAAT and manikin indices of raw approach-avoidance bias (H2a and H2b) and relative approach-avoidance bias (H2c and H2d), separately for physical activity and for sedentary stimuli. Correlation coefficients will be reported alongside 95% confidence intervals and corresponding p-values. The significance level will be set at $\alpha = .05$.	For H2a-d, sensitivity power analysis was conducted using G*Power for a bivariate correlation exact test, assuming a two-tailed $\alpha = .05$, 80% power, and a total sample size of $N = 96$, and indicated a minimum detectable correlation of $r = .28$.	- Significant positive correlations with a 95% confidence interval entirely above $r = .28$ will be interpreted as support for concurrent validity. - Positive correlations with 95% confidence intervals including $r = .28$, irrespective of statistical significance, will be interpreted as inconclusive with respect to associations of at least this magnitude. - Positive correlations with 95% confidence intervals entirely below $r = .28$, irrespective of statistical significance, will be interpreted as indicating that the data are inconsistent with an association of at least this magnitude. - Significant correlations in the opposite direction will be interpreted as indicating inconsistency between tasks, calling into question their comparability.

What is the convergent validity of the KAAT?	Hypotheses: - relative bias towards physical activity derived from the KAAT would be associated with (H3a) device-based total physical activity, (H3b) device-based moderate-to-vigorous physical activity, (H3c) explicit affective attitudes towards physical activity, (H3d) intentions to be physically active, (H3e) valuation of physical effort, and (H3f) automaticity towards physical activity - these associations would be stronger for the KAAT than for the manikin task (H3g-l).		- For H3a–f, separate multiple regressions will be estimated for each outcome variable. Each model will include 7 covariates (age, sex, height, weight, handedness, error rate, and elapsed time) + the KAAT relative bias as the predictor of interest. - For H3g–l, separate multiple regressions will include the same covariates + the KAAT relative bias + the manikin relative bias. Including both biases in the same model allows estimation of their unique (partial) associations with each outcome while accounting for shared variance. Planned contrasts comparing regression coefficients ($\beta_{KAAT} - \beta_{manikin}$) will be conducted within each model to test whether the KAAT relative bias is more strongly associated with each outcome than manikin bias. - To control for multiple comparisons within H3a–f and H3g–l [78], the six hypothesis tests in each set will be treated as a single family and p-values will be adjusted using the Benjamini-Hochberg False Discovery Rate procedure ($q = .05$) [79,80]. Device-based total physical activity will be specified a priori as the primary outcome within each family and will be evaluated without correction.	We conducted sensitivity power analyses in G*Power (F tests; linear multiple regression: fixed model, R^2 increase, 8 predictors, 1 tested predictor) to approximate the minimum detectable regression coefficient (for H3a–f). The smallest detectable incremental effect was $f^2 = 0.084$. This corresponds to approximately $\Delta R^2 = 0.05$–0.08, when assuming plausible full-model R^2 values between 0.10 and 0.40, yielding a detectable semi-partial correlation of approximately $r = 0.23$–0.28. Without significant multicollinearity, this corresponds approximately to the magnitude of the β_{KAAT} coefficient that the analysis is expected to detect with adequate statistical power. We did not estimate a minimum detectable coefficient difference (H3g–l), because G*Power cannot provide a reliable effect size for this contrast	- A significant standardized coefficient β_{KAAT} with a 95% confidence interval entirely above zero will be interpreted as evidence that the KAAT relative bias is positively associated with the corresponding outcome. A significant negative coefficient with a confidence interval entirely below zero will be interpreted as evidence of an association in the opposite direction. - A significant positive planned contrast will be interpreted as evidence that KAAT relative bias is more strongly associated with the corresponding outcome than manikin relative bias. A significant negative contrast will indicate the opposite. - KAAT coefficients with 95% confidence intervals including effects of approximately $\beta_{KAAT} = .23$–$.28$, irrespective of statistical significance, will be interpreted as inconclusive with respect to associations of at least this magnitude. - KAAT coefficients with a 95% confidence interval entirely within the lower limits of the minimum detectable effect (e.g., $-.23$ to $+.23$) will be interpreted as indicating that the data are inconsistent with associations of at least this magnitude. - For secondary outcomes, significant level will be adjusted using the Benjamini-Hochberg false discovery rate procedure ($q = .05$).
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What is the predictive validity of the KAAT?	Hypotheses: - reaction-time-based relative bias towards physical activity stimuli measured with the KAAT would explain variance in device-based total physical activity (H4a) - adding kinematic-based relative biases (i.e., time to peak velocity, x-flips, and opposite displacement) to the model will explain additional variance (H4b) - reaction-time-based relative bias towards physical activity stimuli measured with the KAAT would explain more variance than when assessed with the manikin task (H4c).		For H4a and H4b, a multiple linear regression model will be estimated in three sequential steps. Step 1 = covariates. Step 2 = covariates + KAAT relative bias for physical activity. Step 3: covariates + KAAT relative bias + kinematic-derived biases. - H4a and H4b will be evaluated by examining the change in explained variance (ΔR^2) at each step. Changes in explained variance will be tested using F-tests for nested models. Standardized regression coefficients (β), 95% confidence intervals, and ΔR^2 values will be reported. - For H4c, a separate regression including the same covariates as in Step 2 will be estimated, replacing the KAAT-derived relative bias with the manikin-derived relative bias. The predictive performance of the KAAT-based model (Step 2) and the manikin-based model will then be compared. Because these models are not nested, comparisons will rely on adjusted R^2 and information criteria (AIC and BIC) (81,82) rather than F-tests.	For H4a–b, similar sensitivity power analysis parameters used for H3a–f were applied to assess the smallest detectable increments in explained variance (ΔR^2) - For H4a, predictors were identical to those in H3a–f, yielding a minimal detectable incremental effect size of $f^2 = 0.084$. - For H4b, three predictors (x-flips, opposite displacement, and time to peak velocity) were entered at the final step of the model, resulting in 11 total predictors. The minimum detectable incremental effect size was $f^2 = 0.119$. - Expressed in terms of explained variance, these values correspond to detectable increases in ΔR^2 of approximately 5–8% for H4a and 7–11% for H4b. Sensitivity power analysis is not applicable for H4c.	- A significant increase in explained variance (ΔR^2) after the addition of the predictors will be interpreted as evidence of incremental predictive validity. - If the observed results are below the minimum detectable effect size (approximately $\Delta R^2 = 0.05$–0.08 for H4a and $\Delta R^2 = 0.07$–0.11 for H4b, depending on the full-model R^2), this will be interpreted as limited sensitivity and the need for larger replication studies. - For H4c, lower AIC and BIC values for the KAAT-based model will indicate greater relative support for that model. - Lower AIC and BIC values for the manikin-based model will be interpreted as evidence against the hypothesised predictive advantage of the KAAT - Comparable model fit (i.e., ΔAIC or $\Delta BIC < 2$) will be interpreted as suggesting similar predictive validity of both tasks.
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